CS258

THE UNIVERSITY OF WARWICK

Second Year Examinations: 2017/2018

Database Systems

Time allowed: 2 hours

Answer **FOUR** questions out of the six.

Read carefully the instructions on the answer book and make sure that the particulars required are entered on **each** answer book.

Calculators are not required and not permitted.

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1. (a) What are NULLs? What extra checks may be needed during SQL comparison operations to handle them and why? [3]

   (b) What constraint properties does a PRIMARY KEY possess in SQL? Describe why these constraints are necessary. [4]

   (c) Consider the TAXI schema diagram in Figure 1 overleaf. Write SQL code to create the following views:

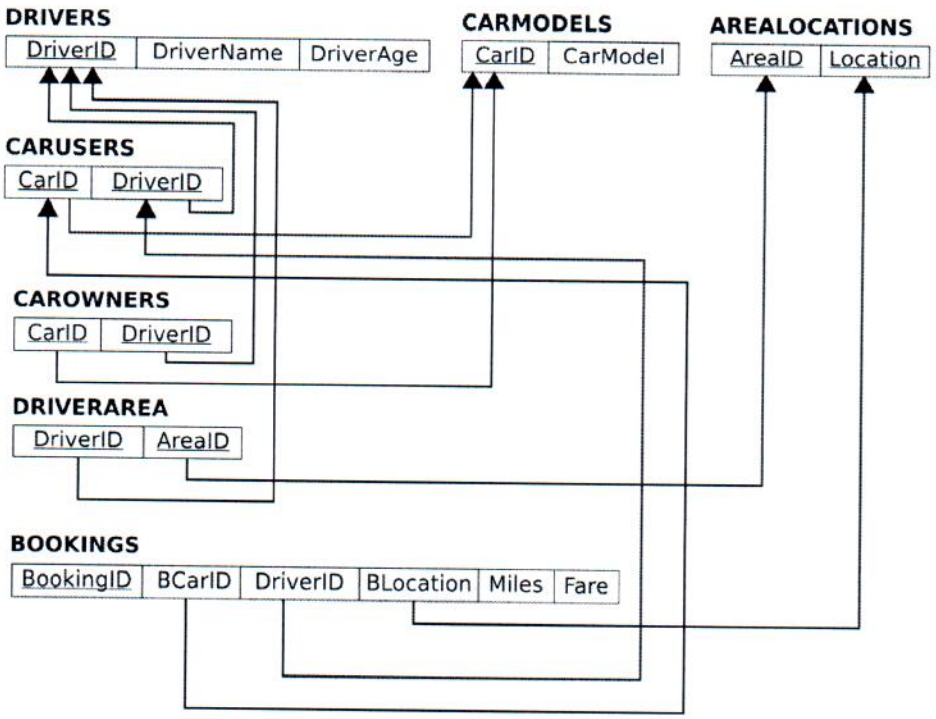
      i. A view named BDRIVERS that provides the following details for every booking: the booking ID and the name of the driver on that booking. [3]

      ii. A view named EFFICIENCY that, for each unique pair of driver ID and location that exists as part of a booking, provides the following details: driver ID, location, and the average fare per mile for that driver ID and location across all bookings. [5]

      iii. A view named COVERAGE that lists the names of all drivers who have had a booking in every location associated with area ID 4. [10]

   Total mark for this question: 25

Continued



The TAXI database stores details about a taxi company's system. The above diagram shows the schema and referential integrity constraints.

CARMODELS stores an ID (CarID, Integer) for every car in the system, along with the car's model (CarModel, VarChar).
DRIVERS stores an ID (DriverID, Integer) for every potential driver in the system, along with their name (DriverName, VarChar) and their age (DriverAge, Integer).
AREALOCATIONS stores an ID (AreaID, Integer) and any locations (Location, VarChar) associated with this ID. Every area in the system is broken up into a collection of locations.
CAROWNERS stores information about which cars drivers own (but do not necessarily use), associating drivers (DriverID, Integer) with cars (CarID, Integer).
CARUSERS stores information about which cars drivers actively use, associating drivers (DriverID, Integer) with cars (CarID, Integer).
DRIVERAREA stores information about which areas (AreaID, Integer) a driver (DriverID, Integer) serves.
BOOKINGS stores information about all taxi bookings that have ever been taken, including the car used (BCarID, Integer), the driver (DriverID, Integer), the location (BLocation, VarChar), the number of miles (Miles, Integer) and the fare collected (Fare, NUMERIC(9,2) or NUMBER(9,2)). This is all associated with an ID (BookingID, Integer).

Figure 1: TAXI Database

Continued

2. (a) Describe the difference between inner joins, left outer joins and cross joins. [3]

   (b) Write the following queries in SQL for the TAXI database in Figure 1 on the previous page:

      i. Retrieve the names of drivers who use a car model that starts with 'Audi'. [4]

      ii. Retrieve the IDs of areas served by at least 8 drivers. [4]

      iii. Retrieve without repetition any car model that has an instance of a car that has been driven over 10000 miles in total across all bookings. [5]

      iv. Retrieve a list of unique pairings of non-matching driver IDs who share the use of at least one car but do not share any areas, with headings named 'Driver1' and 'Driver2' [9]

   Total mark for this question: 25

3. Specify the following queries for the TAXI database (Figure 1 on the previous page) in relational algebra:

   (a) Retrieve a list of car models for all cars used in bookings in the location 'West Birmingham' [5]

   (b) Retrieve locations served by drivers aged between 25 and 35 (inclusive for both). [6]

   (c) Retrieve the names of drivers who do not use any cars that they own. Name the output list 'Owners'. [7]

   (d) Retrieve the names of drivers who own cars that could potentially be used in every area by the current drivers. [7]

   Total mark for this question: 25

Continued

4. (a) What are the components of a relation heading? [2]

   (b) What is the closed world assumption and how is it linked to relations? [4]

   (c) Consider the relations PERSONS(FirstName, Lastname, Age, FavouriteFood) and FOODPRICES(FoodName, Price). Given the data-set below:

FirstName	LastName	Age	FavouriteFood
Bob	Brown	19	IceCream
Jill	Brown	20	Truffles
Jack	Simons	18	Chocolate
Sarah	Matthews	21	Chocolate

(a) PERSONS

FoodName	Price
IceCream	1
Truffles	50
Chocolate	1
Truffles	45

(b) FOODPRICES

   i. Is the set of potential superkeys the same as the set of potential candidate keys for relation PERSONS? Give your reasoning. [3]

   ii. Is the set of potential superkeys the same as the set of potential candidate keys for relation FOODPRICES? Give your reasoning. [3]

   (d) Given the relation schema R(ABCDE), give the candidate key(s) for the following sets of functional dependencies. Explain your answers.

   i. $\{D\} \rightarrow \{A, E\}, \{B\} \rightarrow \{C\}, \{C\} \rightarrow \{E\}, \{A\} \rightarrow \{B\}$ [3]

   ii. $\{B, C\} \rightarrow \{E\}, \{B\} \rightarrow \{D\}, \{A\} \rightarrow \{C\}$ [4]

   iii. $\{D\} \rightarrow \{E\}, \{B\} \rightarrow \{A\}, \{B, C\} \rightarrow \{D\}, \{E\} \rightarrow \{D\}, \{A, D\} \rightarrow \{C\}$ [6]

Total mark for this question: 25

5. Consider the relation schema:

   FILMS(FilmName, ReleaseYear, CastMember, AgeRating, Role, Pay, Curfew)

   where {FilmName, ReleaseYear, CastMember} is the primary key. Assuming the additional dependencies below, show your working to answer the following:

$$\{\text{AgeRating}\} \rightarrow \{\text{Curfew}\}$$
$$\{\text{CastMember}\} \rightarrow \{\text{Role}\}$$
$$\{\text{FilmName, ReleaseYear}\} \rightarrow \{\text{AgeRating}\}$$

   (a) Based on the primary key, determine whether this relation is in 1NF, 2NF, 3NF. [9]

   (b) Perform any normalisation steps required to bring it into 3NF. Show your working and reasoning, including highlighting any integrity constraints. [16]

Total mark for this question: 25

6. (a) Why is it important to use prepared statements? [3]

   (b) What is the difference between horizontal propagation limits and vertical propagation limits in Discretionary Access Control (DAC)? [3]

   (c) A file of 8757 blocks is to be sorted with an available buffer space of 20 blocks. How many passes will be needed in the merge phase of the external sort-merge algorithm? Show all steps of your working. [4]

   (d) Apply the six-step heuristic algebraic optimisation algorithm to the following SQL query (using the TAXI database in Figure 1) to construct an optimised relational algebra query tree:

```
SELECT BookingID, Fare
FROM BOOKINGS, CARMODELS, AREALOCATIONS
WHERE areaID > 4 AND CarModel='Ford'
AND Location = BLocation AND CarID = bCarID
```

   Show your working by converting the SQL query to a relational algebra query tree and drawing the state of the tree before applying the algorithm, and after steps 2, 3, 4 and 5. Discuss what changes you made to the tree at each stage and why. If no changes were made, discuss why not. [15]

Total mark for this question: 25

End